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Analysis of Auditory Evoked Potentials by Magnitude-Squared Coherence*

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ABSTRACT

Evoked potentials are usually analyzed in the time domain (voltage versus time). The most familiar frequency-domain measure, the power spectral density function, displays power as a function of frequency but doesn't distinguish signal power from noise power. The coherence function estimates, for each frequency, the ratio of signal power to total (signal plus noise) power and, thus, indicates the degree to which system output (scalp potential) is determined by input (acoustic stimulus). Coherence ranges from 0 to 1; values above specified critical values can be accepted as demonstrating statistically significant system response. In this paper, we present coherence analysis of human scalp responses to clicks and amplitude-modulated tones. In both cases, this analytic method provides insight into the spectral character of the response (for example, assisting in specifying desirable filter characteristics). Threshold sensitivity is also improved: statistically significant responses can be detected at lower intensity by coherence analysis than by inspection of time-domain waveforms.

Auditory evoked potentials (AEPs) have usually been averaged and analyzed in the time domain, i.e., as waveforms representing scalp voltage as a function of time. Amplitudes, latencies, and thresholds—the latter most often estimated by inspection—have been the dependent variables of traditional interest. AEPs can also be analyzed in the frequency domain. Beyond its inherent descriptive interest, spectral analysis, by identifying the frequencies prominent in a particular AEP type, assists in the selection of response filter passbands (Fridman, John, Bergelson, Kaiser, & Baird, 1982). Comparison of the power spectrum measured during signal presentation (containing system response plus system noise) to that measured in a no-stimulus condition (noise only) permits a judgment of the presence or absence of a response. However, comparison of power spectra is difficult when a low-amplitude response is present, and is not easily amenable to statistical inference.

The goal of this paper is to present and illustrate the use of the magnitude-squared coherence function (γ^2) as an alternative to "simple" spectral analysis of AEPs. Although the examples in this paper use AEPs, coherence analysis is equally applicable to evoked potentials from any sensory modality.

γ^2 is a measure of the degree to which system output (such as an AEP) is determined by a specified input, as a function of frequency (Bendat & Piersol, 1986). For a linear system, coherence measures, for each frequency of interest, the proportion of response power attributable to a given stimulus; it is thus a "signal to signal plus noise" power ratio, varying from 0 to 1. Multiple response spectral measures are required for a single "smoothed" and valid coherence estimate; these can be obtained by breaking up a single evoked potential average into multiple subaverages (e.g., 16 subaverages of 256 responses each instead of a single average of 4096), or by combining spectral estimates for adjacent frequencies (at the expense of frequency resolution). These two methods—smoothing across subaverages or frequencies—can also be used together to further reduce variance. Depending on the degree of smoothing, critical values can be determined for any desired significance level; coherence estimates above these values indicate statistically significant response.

The use of periodic signals, as is necessary in evoked potential work, poses some problems: the measured coherence values no longer are valid estimates of linear relationship between stimulus and response. However, the technique is still useful, since our main interest is not in actually measuring coherence per se, but in determining whether a response is present. An estimated coherence value above the appropriate critical value does this, whether the response is linear or not. An advantage of using periodic stimuli is that coherence can be very rapidly calculated using only the spectra of the subaverages and the grand average (mean) response: for a given frequency, coherence is then equal to the power of the mean response

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divided by the mean power (of the subaverages).

The basic principle is that during the averaging process, signal (phase-locked to the stimulus) remains while noise (uncorrelated to the stimulus) tends to cancel out. We periodically interrupt this process to measure subaverages in addition to the grand average response, then calculate power spectra for each. For a frequency containing only signal, the powers in the subaverages and the grand average will all be the same, and coherence will be 1. For any frequency containing only noise, the power in the grand average will be much less than the mean power in the subaverages, and estimated coherence will be much less than 1, tending toward 0 as the number of subaverages increases. When signal and noise are mixed, coherence will depend on the signal to noise ratio.

The following section, "Calculation of Coherence Estimates," is mathematically intuitive and nonrigorous, requiring no calculus, and should be understandable by both clinicians and investigators working with evoked potentials. However, readers willing to accept the preceding assertions on faith may skip directly to the "Methods" section.

CALCULATION OF COHERENCE ESTIMATES

Frequency Domain Analysis

Table 1 lists and briefly defines abbreviations and symbols used. The "system" of interest is a human or animal subject receiving an acoustic input, x(t), and producing a scalp voltage output y(t)—y(t) can be considered to be the sum of system response to the input, S(t), and random noise produced by the system, N(t).

Fourier analysis produces complex frequency domain representations of the input and output signals, X(f) and Y(f); for each of these, real and imaginary (orthogonal) components are calculated for each frequency. Amplitude A(f) and phase θ(f) can be derived by well-known trigonometric relationships (Fig. 1 compares the polar and complex representations of X(f)):

$$A(f) = \sqrt{Re[X(f)]^2 + Im[X(f)]^2} \quad (1)$$

$$\theta(f) = \tan^{-1}\left(\frac{Im[X(f)]}{Re[X(f)]}\right) \quad (2)$$

A(f) is sometimes indicated as $|X(f)|$, i.e., the "absolute value" of the complex number X(f) equals the length of its vector.

The power spectral density function, $G_{xx}(f)$, the frequency domain display most familiar to evoked potential workers, is a real-valued function equal to the square of A(f). For the input x(t):

$$G_{xx}(f) \doteq A_x^2(f) = |X(f)|^2 \quad (3)$$

Note that $G_{xx}(f)$ can also be obtained by complex multiplication. X*(f), the complex conjugate of X(f), is a function whose real components are identical to those of X(f), but whose imaginary components have reversed signs (the phases are thus also reversed). Since the phase of the product of two complex numbers is the sum of the phases of the original numbers, X(f)X*(f) has zero phase [θ(f) − θ(f)], and is thus real-valued. Therefore, $G_{xx}(f)$ = X(f)X*(f).

Table 1. Abbreviations and symbols.

A (f)	Amplitude of Fourier component as a function of frequency f
cos	Cosine
f	Frequency (f_c = carrier frequency; f_m = modulating frequency)
G_{xx} (f)	Power spectral density function of x (t)
G_{xy} (f)	Cross spectral density function relating x (t) and y (t) as a function of frequency f
i, j	Subscripts indicating one of a series of measurements
Im [ ]	Imaginary component of [ ]
ℓ	Number of adjacent frequencies over which spectral estimates are averaged prior to calculation of coherence
m	Number of adjacent frequencies over which multiple coherence estimates are averaged
n	Number of measurements averaged in the time domain
N (t)	Portion of system response y (t) attributable to noise
p	Probability
PC	Phase coherence
q	Number of measurements averaged in the frequency domain
Re [ ]	Real component of [ ]
S (t)	Portion of system response y (t) attributable to input x (t)
sin	Sine
t	Time
tan	Tangent
x (t)	Input signal (time domain)
X (f)	Fourier transform (frequency domain) of input signal
y (t)	System output (time domain)
Y (f)	Fourier transform (frequency domain) of system output
z	Number of standard deviations encompassing a given proportion of a normal distribution
α	Desired level of significance for statistical inference
Δ	A small change in some quantity
γ^2 (f)	Magnitude-squared coherence relating x (t) and y (t) as a function of frequency f
θ (f)	Phase angle as a function of frequency f
Σ	Sum
($\bar{\ }$)	Superscript indicating an averaged measurement
($\hat{\ }$)	Superscript indicating an estimate
(*)	Indicates a complex conjugate (imaginary component multiplied by −1); thus, X = Re [X] + Im [X]; X* = Re [X] − Im [X]
$\| \ \|$	Absolute value

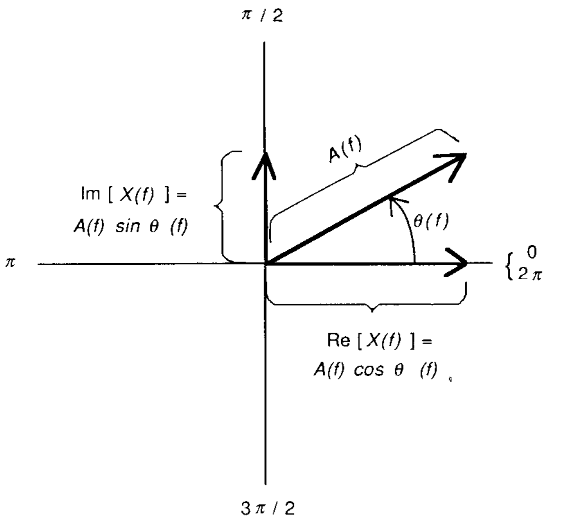

Figure 1. The amplitude and phase of a single frequency component (f) of a signal X(f) are shown in polar coordinates. Note that one can equivalently represent this vector by its Cartesian coordinates, the real and imaginary (orthogonal) parts of X(f).

We now consider the cross-spectral density function $G_{xy}(f)$, which relates x(t), the system input, to y(t), the system output, as a function of frequency f:

$$G_{xy}(f) = X^*(f)Y(f) \quad (4)$$

The amplitude of $G_{xy}(f)$ is the product of the amplitudes of X(f) and Y(f), while its phase will be the difference between output and input phase, i.e., the phase lead or lag introduced by the system. $G_{xy}(f)$ can be normalized by division by the square root of the product of the input and output power spectra, so that its real and imaginary components vary between −1 and +1. This normalized cross-spectrum is the complex coherence function, $\gamma_{xy}(f)$.

Coherence

γ^2_{xy} is the magnitude-squared coherence; assuming a linear system:

$$\gamma^2_{xy}(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} \quad (5)$$

which varies from 0 to 1 and is real-valued. $\gamma^2(f)$ estimates the proportion of the power in y(t) which is attributable to input x(t), for frequency f. Thus,

$$\gamma^2(f) = \frac{|S(f)|^2}{|Y(f)|^2} = \frac{|S(f)|^2}{|S(f)|^2 + |N(f)|^2} \quad (6)$$

Algebraically manipulating this in terms of the signal to noise ratio, S/N (omitting the "f" for simplicity), yields

$$\gamma^2 = \frac{(S/N)^2}{(S/N)^2 + 1} \quad (7)$$

and

$$S/N = \sqrt{\frac{\gamma^2}{1 - \gamma^2}} \quad (8)$$

Expanding equation 5 yields an interesting result for the case in which all the necessary functions are estimated from a single pair of time segments $\hat{x}(t)$ and $\hat{y}(t)$. The superscript (ˆ) indicates an estimate of the "true" function based on a finite data sampling. Based on equations 3 and 4:

$$\hat{\gamma}^2(f) = \frac{|\hat{X}^*(f)\hat{Y}(f)|^2}{|\hat{X}(f)|^2\,|\hat{Y}(f)|^2} = \frac{|\hat{X}^*(f)|^2\,|\hat{Y}(f)|^2}{|\hat{X}(f)|^2\,|\hat{Y}(f)|^2} = 1 \quad (9)$$

Since $\hat{X}^*(f)$ and $\hat{X}(f)$ have identical amplitudes, their powers are also identical, and $\hat{\gamma}^2 = 1$. Coherence estimates from a single pair of samples will always be 1, regardless of the true coherence. This is analogous to estimating a correlation coefficient (e.g., for height and weight) from measurements of one individual; the result will be (spuriously) equal to 1.

For this reason, γ^2 estimates must always be smoothed, based on multiple independent samples. Most often, this is done by collecting multiple data segments for both x(t) and y(t), then averaging (for each frequency) the spectral estimates from the q segments, to yield the best overall estimates for $G_{xy}(f)$, $G_{xx}(f)$, and $G_{yy}(f)$:

$$\hat{G}_{xy}(f) = \frac{1}{q}\sum X_i^*(f)Y_i(f) \quad (10)$$

$$\hat{G}_{xx}(f) = \frac{1}{q}\sum |X_i(f)|^2 \quad (11)$$

$$\hat{G}_{yy}(f) = \frac{1}{q}\sum |Y_i(f)|^2 \quad (12)$$

$$\hat{\gamma}^2_{xy}(f) = \frac{|\hat{G}_{xy}|^2}{\hat{G}_{xx}(f)\,\hat{G}_{yy}(f)} \quad (13)$$

The averaging in equation 10 operates separately on the real and imaginary parts of $X_i^*(f)Y_i(f)$, not on amplitudes and phases per se. Thus, the real part of $\hat{G}_{xy}(f)$ is the average of the real parts of cross-spectra estimated from q separate segments of data. If there is no coherence between x(t) and y(t) at frequency f, the phases of the q separate estimates of $G_{xy}(f)$ will be randomly distributed, and both real and imaginary components will be as often negative as positive. Their averages and the magnitude of $\hat{G}_{xy}(f)$ will tend toward 0 as q increases. Conversely, the denominator of equation 13 is equal to the product of the mean powers of the q estimates of x(t) and y(t) and will not be reduced by averaging over data segments. Thus, when true coherence $\gamma^2 = 0$, estimated coherence $\hat{\gamma}^2$ will approach 0 and the variance of $\hat{\gamma}^2$ will decrease as q increases.

The spectral estimates can also be averaged across adjacent frequencies to reduce variance. For example, an FFT yielding 128 frequencies can be smoothed by calculating complex average values for groups of 4 frequencies, yielding a function with only 32 frequency values. If ℓ equals the number of frequencies in each group, and f now represents the center frequency for each group,

$$\hat{G}_{xy}(f) = \frac{1}{\ell} \sum X^*(f_j)Y(f_j) \tag{14}$$

$\hat{G}_{xx}(f)$ and $\hat{G}_{yy}(f)$ are similarly smoothed. Frequency averaging and segment averaging can be used together, with the net reduction in variance of $\hat{\gamma}^2$ related to the product $q\ell$.

Our main interest in evoked potential work is in rejecting the null hypothesis $\gamma^2 = 0$, i.e., confirming the presence of a response. Critical values of $\hat{\gamma}^2$ for $\gamma^2 = 0$ (levels above which a response can be accepted) have been presented by various authors, based on theoretical calculations (Amos & Koopmans, 1963; Carter, Knapp, & Nuttall, 1973) and empirical simulations (Benignus, 1969). Carter et al. propose that for $q\ell > 32$, and γ^2 (true coherence) = 0, the bias of $\hat{\gamma}^2$ will be $1/q\ell$, and the variance $1/(q\ell)^2$. If the critical value for $\hat{\gamma}^2$ is equal to bias plus z standard deviations (determined by desired statistical significance level, α), then

$$\text{critical value}_\alpha\ (\hat{\gamma}^2) = \frac{1 + z}{q\ell} \tag{15}$$

(This calculation assumes that $\hat{\gamma}^2$ is normally distributed; as will be seen, this assumption is unjustified and produces slightly too-liberal critical values.)

Recall from equation 7 the relation between γ^2 and the S/N ratio. For S/N $\ll 1$, as would be typical of most short-latency AEPs, γ^2 is approximately $(S/N)^2$. If S/N = 0.1 (i.e., −20 dB), $\gamma^2 = 0.01$. From equation 15, for a one-tailed test at a 99% confidence level ($\alpha = 0.01$), we would require $q \geq 333$ to reject the null hypothesis for a measured coherence value this low (assume for the moment that $\ell = 1$ and all smoothing is via segment averaging). At least for software-based systems, this would make inordinate demands on processing time and/or memory, and it is usually more practical to perform some time domain averaging prior to spectral analysis.

If each of q data segments is the result of averaging n separate samples, the resultant "subaverages" will have S/N ratios improved by a factor of $\sqrt{n}$, and γ^2 will increase approximately as n. Conversely, the required degree of smoothing will decrease by a similar factor, since (for large $q\ell$) the critical values of $\hat{\gamma}^2$ are inversely proportional to $q\ell$. For a fixed total number of samples (= qn), there is a trade-off between q and n. For high q and low n, $\hat{\gamma}^2$ and its critical value will be high. For low q and high n, both will be low.

Smoothing across frequencies ($\ell > 1$) decreases $\hat{\gamma}^2$ variance without requiring increased data collection time, but at the expense (often very acceptable) of decreased spectral resolution. Also, if there are rapid phase shifts in the system's transfer function for small frequency changes, cross-spectral values for adjacent frequencies will be out of phase and may tend to cancel one another, even if coherence is high for each.

We have so far ignored the nature of the stimulus x(t). We must assume that x(t) is a stationary time series, i.e., its average properties over successive segments of time do not change; the most appropriate stimulus for coherence analysis is Gaussian random noise. For such stimuli, $\gamma^2_{xy}(f)$ reflects the degree to which x(t) and y(t) are related by a linear process. It is often convenient to use periodic, deterministic inputs such as pseudorandom noise, particularly when extensive time domain averaging must be done to improve S/N ratios. Maki (1986) has shown that such stimuli permit nonlinearities present in the system to contribute to $\hat{\gamma}^2_{xy}(f)$. For example, harmonics generated in response to a given input frequency can result in apparently significant coherence for frequencies not present in the input. He suggests the use of pseudorandom stimuli composed only of prime-numbered harmonics, none of which are integer multiples of other components, to eliminate at least some kinds of nonlinearities from $\hat{\gamma}^2_{xy}(f)$. Even-order nonlinearities, such as are introduced by rectification, can be avoided by using inverse-repeat stimuli (Swerup, 1978); such stimuli have energy only at odd multiples of their fundamental frequency.

When a periodic stimulus is used (this would include not only pseudorandom noise, but any stimuli time-locked to the data collection period, such as clicks, tonebursts, etc.), and spectral estimates are averaged only across data segments, it can be shown that $\hat{\gamma}^2_{xy}(f)$ depends only upon output data. Since X(f) is invariant,

$$\gamma^2(f) = \frac{|X^*(f)|^2|\hat{Y}(f)|^2}{|X(f)|^2\hat{G}_{yy}(f)} = \frac{|\hat{Y}(f)|^2}{\hat{G}_{yy}(f)} = \frac{\left|\frac{1}{q}\sum Y_i(f)\right|^2}{\frac{1}{q}\sum |Y_i(f)|^2} \tag{16}$$

Restated, coherence in this case is equal to the power of the grand average response divided by the average power of the subaverages. When spectral estimates are averaged across adjacent frequencies ($\ell > 1$), however, this simplification is not applicable. Since the amplitudes and phases of adjacent frequencies in the stimulus may be quite different, the cross-spectra must be separately estimated for all frequencies prior to averaging.

Our software used algorithms based on equations 10 to 14. However, for any frequency not represented in the stimulus, this would lead to an undefined coherence since zeroes would appear in both numerator and denominator of equation 13. For these cases, equation 16 was used.

"Phase Coherence" and Other Measures

We must distinguish the magnitude-squared coherence $[\gamma^2(f)]$ from the "phase coherence" (PC) statistic which has recently been introduced to the AEP literature, for use in analysis of steady state evoked potentials (Jerger, Chmiel, Frost, & Coker, 1986; Picton, Vajsar, Rodriguez, & Campbell, 1987b; Stapells, Makeig, & Galambos, 1987). As for γ^2, PC analysis breaks up the data collection period into q subaverages, but instead of then carrying out complex averaging of the response or cross-spectrum vectors, the PC approach considers only the phases of each subaverage for a given frequency. These phase angles are projected onto a unit circle, and their sines and cosines are separately averaged. The length of the resulting vector (ranging from 0–1) is a measure of relative phase aggregation:

$$PC = \sqrt{\left(\frac{1}{q}\sum \cos \theta_i\right)^2 + \left(\frac{1}{q}\sum \sin \theta_i\right)^2} \qquad (17)$$

Compare this to $\hat{\gamma}^2$, for the special case of an invariant input (equation 16):

$$\hat{\gamma}^2 = \frac{\left|\frac{1}{q}\sum Y_i(f)\right|^2}{\frac{1}{q}\sum |Y_i(f)|^2} = \frac{\left(\frac{1}{q}\sum A_i \cos \theta_i\right)^2 + \left(\frac{1}{q}\sum A_i \sin \theta_i\right)^2}{\frac{1}{q}\sum A_i^2} \qquad (18)$$

If we ignore amplitudes by setting all $A_i = 1$, $\hat{\gamma}^2 = PC^2$. Thus, the PC statistic is analogous to the square root of $\hat{\gamma}^2$, but disregards amplitude information. To the extent that AEPs result from phase reordering of EEG energy rather than the addition of new energy at specific frequencies, this is reasonable.

PC is identical to Goldberg and Brown's (1969) "vector strength," offered as a measure of phase-locking of single-unit discharges, and to the statistic $\bar{R}$ used in the Raleigh test of circular variance (see Mardia, 1972, Appendix 2.5, for a table of critical values for different numbers of subaverages, q). Fridman et al (1984) used a statistic they called "synchrony measure" (equal to PC^2) for objective detection of auditory brain stem responses.

Jervis, Nichols, Johson, Allen, and Hudson (1983) claim, on theoretical grounds, that PC (the Raleigh test) is less powerful than a statistic incorporating both amplitude and phase information, when the AEP actually adds energy to the EEG, rather than simply causing phase aggregation. Their "modified Raleigh" test statistic (nearly identical to $\hat{\gamma}^2$) proved superior to the Raleigh test in detecting long-latency AEPs.

Fridman et al (1982) computed power spectra for multiple subaverages and for a grand average response, then subtracted the spectrum of the grand average from the average spectrum of the subaverages. The difference spectrum value was large for frequencies not contributing to the ABR, and small for those contributing to the response. Their difference score (in dB), in fact, is equivalent to $10 \log_{10}\left(\frac{1}{\hat{\gamma}^2}\right)$, as can be seen from equation 16.

Sayers, Beagley, and Riha (1979) assessed phase aggregation using a simple "rotating Chi-square" test. The unit circle was divided into 24 bins of 15° width. For each of the 12 lines bisecting this circle, a Chi-square test determined whether there was a significant clustering of response phases in one or the other half-circles. Jervis et al (1983) describe an improved version of this test.

Picton, Skinner, Champagne, Kellett, and Maiste (1987) describe the use of the Hotelling T^2 test (Hotelling, 1931) to calculate a "confidence ellipse" for a set of vectors such as those shown in Figure 3: if the origin (0,0) is not included, the mean vector can be considered to represent a statistically significant response. Like γ^2 and Jervis' (1983) "modified Raleigh test," the T^2 test considers both magnitude and phase.

METHOD

Subjects

Subjects were young adults (21–33 years old) with normal hearing (15 dB HL or better for octave-interval pure tones from 250–8000 Hz, ANSI 1970). Otoscopic examinations and otologic histories were unremarkable. This paper presents results for individual subjects, but these were, for each experiment, representative of at least three subjects.

Stimuli

Two types of stimuli (clicks and amplitude-modulated tones) were digitally synthesized using a 256-point array (for details, see Wilson & Dobie, 1987), which was repetitively output at a D/A rate of 5.12 KHz. Thus, the interval between points was about 195.3 μsec, and the period of the entire stimulus waveform was 50 msec (256 × 0.1953); the fundamental frequency (f_0) of the stimulus was 20 Hz. A 500-Hz tone 100% amplitude-modulated at 40 Hz (AM tone) was synthesized by adding the 23rd, 25th, and 27th harmonics (= 460, 500, and 540 Hz, for f_0 = 20 Hz) in appropiate amplitude (1:2:1) and phase relationships. Spectral analysis of the transducer output showed no other spectral components within 70 dB of these three. A click (20/sec) was generated by simply outputting a positive pulse at the beginning of each cycle; alternating-polarity clicks (40/sec) were produced by adding a negative pulse at the midpoint of the cycle.

Both the AM tone and the alternating-polarity click are so-called inverse-repeat stimuli, containing energy only at odd harmonics of the fundamental frequency. (Consider, e.g., the 3rd harmonic. While phase at the end of the stimulus period will be the same as at the beginning, phase at the midpoint will be 180° opposite, 1½ cycles having been completed. Its waveform in the second half of the stimulus period is a mirror image of its first-half waveform, hence "inverse-repeat.") Such stimuli have the interesting property of permitting, in the frequency domain, separation of linear response components from even-order nonlinearities. For AEPs, stimulus artifact, cochlear microphonic, and frequency-following response (FFR) are at least primarily linear, and will contribute energy (and coherence) at the (odd) stimulus frequencies. Response coherence at even frequencies must represent nonlinearity introduced, for example, by rectification (this is presumably how the envelope of an AM tone becomes represented in auditory nerve discharge patterns). Auditory brain stem response (ABR) is itself primarily an even-order nonlinearity, since there is little difference between ABRs to condensation and rarefaction clicks. The common practice of adding responses to clicks of opposite polarity is equivalent to extracting only the even-order nonlinearities while suppressing linear (and odd-order nonlinear) response components (artifact, CM, FFR).

Stimuli were low-pass filtered (24 dB/octave) above 2 kHz, amplified, attenuated, and presented through a TDH-49 earphone in a MX-41/AR supra-aural cushion. Intensity is specified both in decibels sensation level (SL) and in either SPL (for the AM tone) or peSPL (for clicks). Naturally, SL-SPL differences varied slightly across subjects.

Recording and Analysis

Subjects were tested reclining, in a double-walled, sound-treated booth. They were instructed to remain alert but relaxed, and were spoken with by intercom at least every 5 minutes. Gold cup electrodes were attached to alcohol-cleansed skin at the vertex (positive), ipsilateral mastoid (negative), and contralateral mastoid (ground). Electrode impedances were less than 2 kΩ at 30 Hz. Scalp potentials were preamplified (10^5), sampled digitally at 5.12 kHz, and averaged. Preamplifier (Grass P511K) response passband (6 dB/octave) was 30 to 3000 Hz. While the low-pass setting (3 kHz) was greater than the Nyquist frequency (2.56 kHz), the next lower choice was 1 kHz, which would possibly have attenuated response frequencies of interest (ABR spectra roll off rapidly above about 1500 Hz). Time domain responses (except in Fig. 3) are usually presented as replicated averages (typically 2048 sweeps were averaged for each). Coherence functions were obtained as described in the "Calculation" section, with q subaverages, each of which was the average of n responses, and were sometimes further smoothed across groups of ℓ adjacent frequencies. Typically, q = 16 and n = 256, for a total (qn) of 4096 responses; this required about 3.5 minutes of recording time per coherence function (replicated time domain averages were simultaneously acquired). Critical values for $\hat{\gamma}^2$, from the literature, were tested by collecting data from human subjects under "no stimulus" conditions (where, by definition, $\gamma^2 = 0$) and obtaining empirical distributions of $\hat{\gamma}^2$ for various values of q, ℓ and α.

RESULTS

Distribution of Coherence Estimates when True Coherence = 0

For each of several values of q (4, 8, 16, 32, 64), eight coherence functions were calculated using scalp potentials from a subject receiving no stimulus. Each function contained coherence values for 127 frequencies, yielding 1016 (8 × 127) coherence estimates ($\hat{\gamma}^2$) for true coherence ($\hat{\gamma}^2$) = 0. Figure 2a shows the 99th percentile values for this distribution plotted with the theoretical distribution of Amos and Koopmans (1963) and the critical values estimated from Carter et al (1973), assuming a normal distribution (equation 15). Our data agree well with Amos and Koopmans, but the simple estimates based on the Carter et al mean and variance (for q ≤ 32) appear to be slightly too liberal. Figure 2b compares our 97.5th percentile data with the sources mentioned above, and also with the data of Benignus (1969), who carried out a Monte Carlo simulation similar to ours. Both sets of empirical data agree well with the Amos and Koopmans distribution.

We repeated this experiment, varying both q (as above) and ℓ (1, 2, 4, 8, 16). The results are not shown, but confirmed that the 99th and 97.5th percentile points in these distributions fit the Amos and Koopmans distributions very well, based on the ql product. For example, if $q\ell$ = 8 and ℓ = 4, the 99th percentile was about 0.14, as for the case where q = 32 and ℓ = 1. For subsequent experiments, we used critical values from Amos and Koopmans (Table 2). Some of the $q\ell$ values of interest to us are not tabulated in Amos and Koopmans and had to

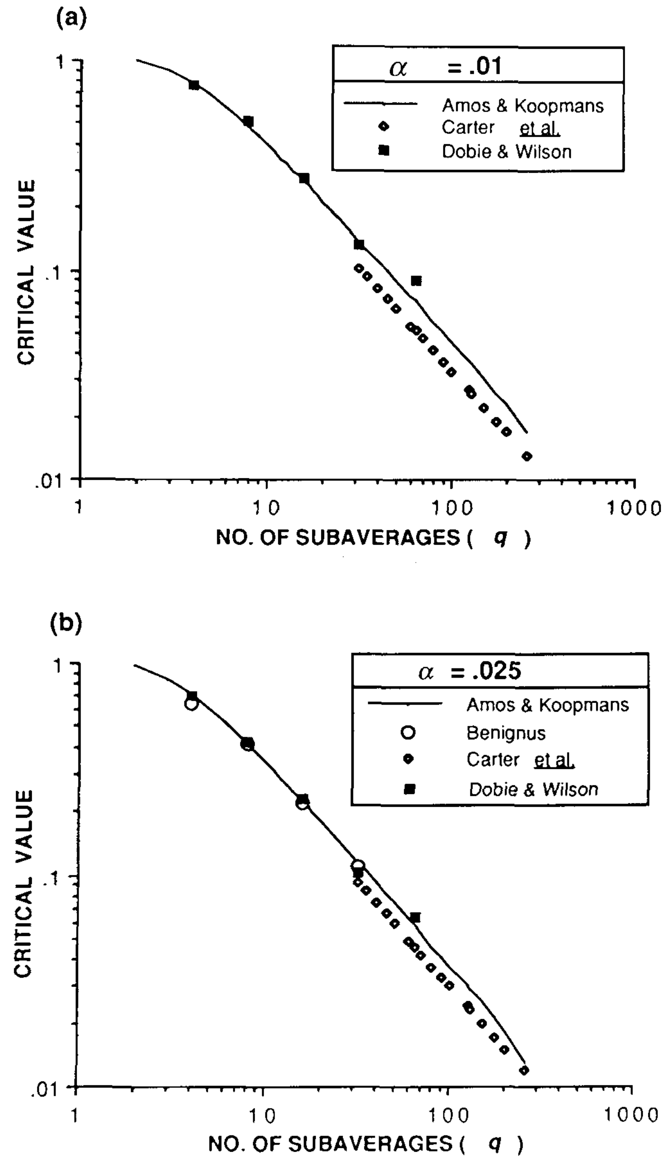

Figure 2. (a), Values (99th percentile) for coherence estimates obtained under no-stimulus conditions are plotted as a function of the number of subaverages (*filled squares*). Critical values for α = 0.01 using the bias and variance estimates of Carter et al (1973), and assuming normal distribution, are also shown (*open diamonds*). The continuous line represents the 99th percentile of coherence estimates for zero true coherence, according to Amos and Koopmans (1963). (b), The same data sources, as well as simulations by Benignus (1969) (*open circles*), for α = 0.025.

be obtained by log-linear interpolation (for example, they included q = 30 and 35, but not q = 32). Note also that their tables show distributions of $|\hat{\gamma}|$ rather than $\hat{\gamma}^2$; the values in Table 2 thus have been squared.

Responses to an AM Tone

Figure 3 shows in some detail how an actual coherence estimate (for one frequency) is obtained. The stimulus was the AM tone (f_c = 500 Hz; f_m = 40 Hz). At low to medium intensities, the primary scalp response follows the stimulus envelope: this is the amplitude-modulated following response (AMFR) described by Kuwada, Batra, and Maher (1986). Like the "40-Hz" response (Galambos, Makeig, & Talmachoff, 1981), it is maximal for modulation frequencies around 40 Hz, but it is less sensitive than response to 40 Hz transients (Palaskas, Wilson, & Dobie, 1988). The stimulus intensity in Figure 3 was 30 dB SL (58 dB SPL), just at the visual detection level of AMFR for this subject. The upper left portion of the figure (a) shows the q (=16) separate subaverages, with the grand average waveform shown below (b). The 40 Hz response appears as two cycles in this 50 msec interval. The upper right part of the figure (c) shows the 40 Hz Fourier components of the q subaverages, and the 40 Hz component of the grand average. The distribution of the powers of the q subaverages is shown in (d). Since a periodic stimulus was used, coherence can be calculated as the power of the mean response divided by the mean power of the subaverages (equation 16); these two power values are shown as scalars in Figure 3d. The result ($\hat{\gamma}^2$ = 0.472) far exceeds the 0.01

Table 2. Critical values for $\hat{\gamma}^2$ (when γ^2 = 0), from Amos and Koopmans (1963).

	Critical Value for Desired α Level (One-Tailed)		
qℓ	0.01	0.025	0.05
4	0.792	0.714	0.632
8	0.483	0.410	0.354
16	0.266	0.219	0.183
32*	0.139	0.114	0.094
64*	0.072	0.058	0.047
128*	0.036	0.029	0.023

* *By interpolation (see text).*

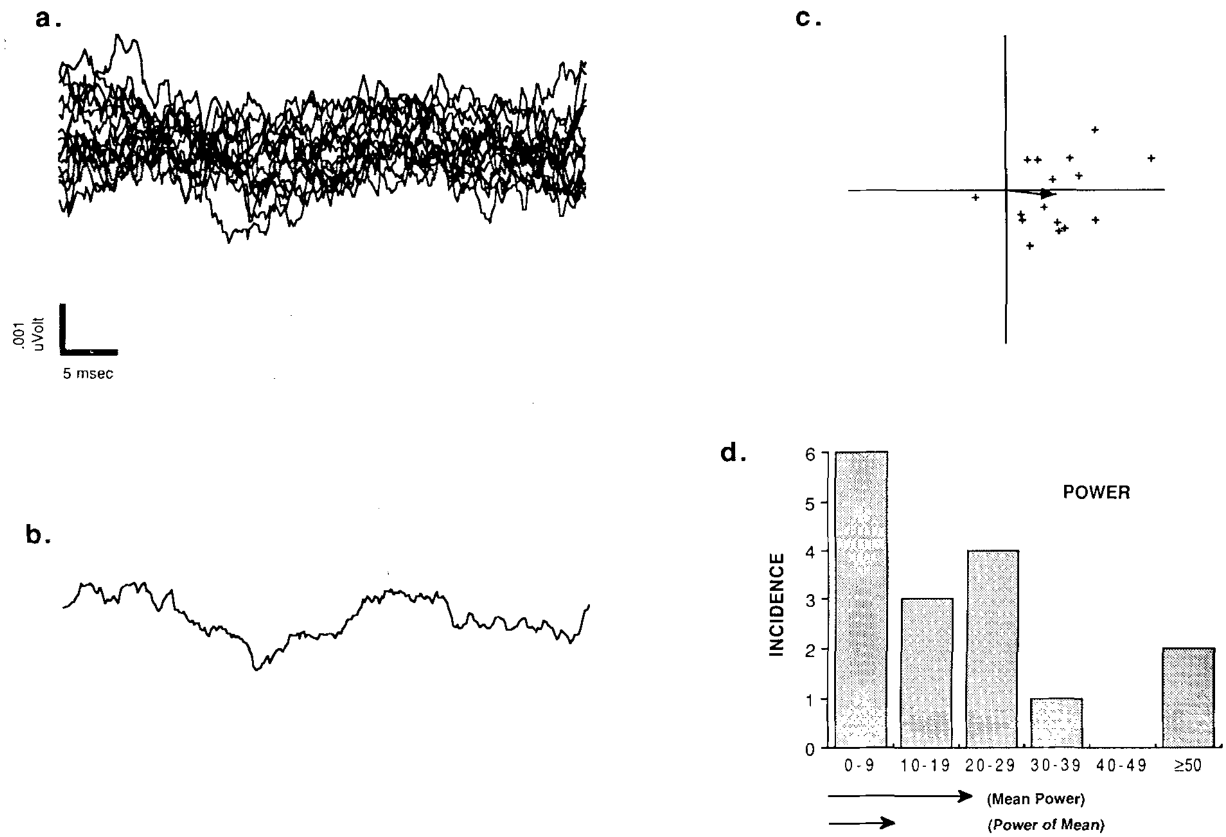

Figure 3. (a), Sixteen "subaverage" scalp responses (n = 256) to an AM tone (30 dB SL), superimposed. (b), The grand average of the waveforms in (a) shows an apparent 40-Hz component (2 cycles within the 50 msec response period). (c), The 40 Hz components of each of the 16 subaverage responses, obtained by FFT, are plotted in polar coordinates ("plus" signs), along with the 40 Hz component of the grand average (vector with arrowhead). (d), The histogram shows the distribution of 40-Hz power for the subaverages; the arrows (scalars) show the mean power of the subaverage responses and the power of the grand average. The ratio of the latter to the former is the estimated coherence.

critical value for q = 16 (Table 2), and indicates with more than 99% certainty that a response was present.

Figure 4 shows time domain response and the coherence functions for the same stimulus (a) at higher intensity (60 dB SL; 85 dB SPL). The 40 Hz envelope response is clearly evident, as is a 500 Hz FFR (Fig. 4b). Cross-correlation of the time domain response with the AM tone stimulus revealed a response latency of 5.96 msec, consistent with an FFR of brain stem origin. The coherence function (Fig. 4d) demonstrates statistically significant coherence for several frequencies: 40 Hz, its 2nd and 3rd harmonics, 500 Hz, the 460 and 540 Hz sidebands also present in the stimulus, and a few frequencies around 1000 Hz. The latter group of frequencies could be considered as harmonics of the stimulus components, but note that they are also present in the envelope of the stimulus (Fig. 4c; obtained by half-wave rectification, as at the hair cell afferent synapse). Rectification is an even-order nonlinearity; for an "inverse-repeat" stimulus containing only odd harmonics, like our AM tone, the new frequencies introduced by rectification are all even harmonics of the fundamental.

Responses to 40 Hz Alternating-Polarity Clicks

Figure 5 shows a series of time domain responses and coherence functions for 40 Hz alternating-polarity clicks (note that this too is an inverse-repeat stimulus). At 70 dB SL one can see stimulus artifact, ABR, and an apparent myogenic response. The coherence function displays statistically significant coherence across the frequency range, especially below 500 Hz and above 1000 Hz. In general, low coherence values alternate with high values. For example, 20 Hz coherence (the first harmonic) is about 0.17, while 40 Hz coherence (second harmonic) is 0.89. Below 460 Hz, all the significant coherences are for even harmonics; above 1560 Hz, only odd harmonics yield significant coherence. Since the stimulus is of the inverse-repeat (odd harmonic) type, most or all of the significant coherence above 1560 Hz is probably attributable to the large stimulus artifact present at this high intensity level.

At 50 dB SL, stimulus artifact is absent, and there is no significant odd-harmonic coherence. There is strong, even-harmonic coherence up to about 750 Hz, with a few values above the critical level for 1000 to 2500 Hz. Any of these isolated peaks could be spurious; since 127 coherence values are computed for each coherence function, there is a 0.72 probability that at least one will exceed the $p < 0.01$ critical value by chance alone $(1 - 0.99^{127})$. Note that 40 Hz coherence is statistically significant down to 10 dB SL, while visual detection of replicable time-domain response fails at about 30 dB SL in this case.

Figure 6 shows 40-Hz coherence as a function of inten-

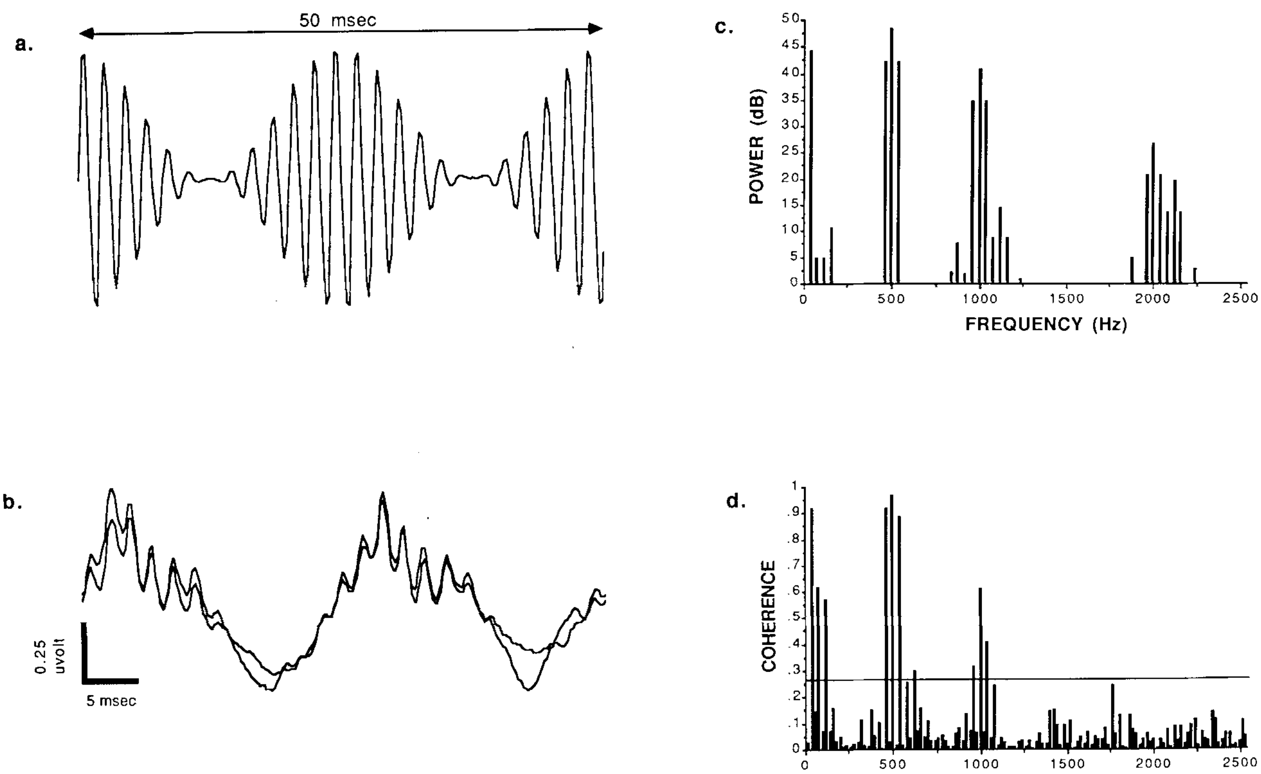

Figure 4. (a), AM tone; f_c = 500 Hz, f_m = 40 Hz. (b), Replicated scalp response showing responses to both f_c and f_m. (c), The envelope of the stimulus was obtained by half-wave rectification; the amplitude spectrum of that envelope is shown here. (d), The coherence function, calculated from the same scalp potential data in (b), shows peaks corresponding to the half-wave rectified stimulus, as well as harmonics of the principal envelope frequency (see text). The critical value at the 0.01 level of significance is indicated by the horizontal line at $\hat{\gamma}^2 = 0.266$ (see Table 2).

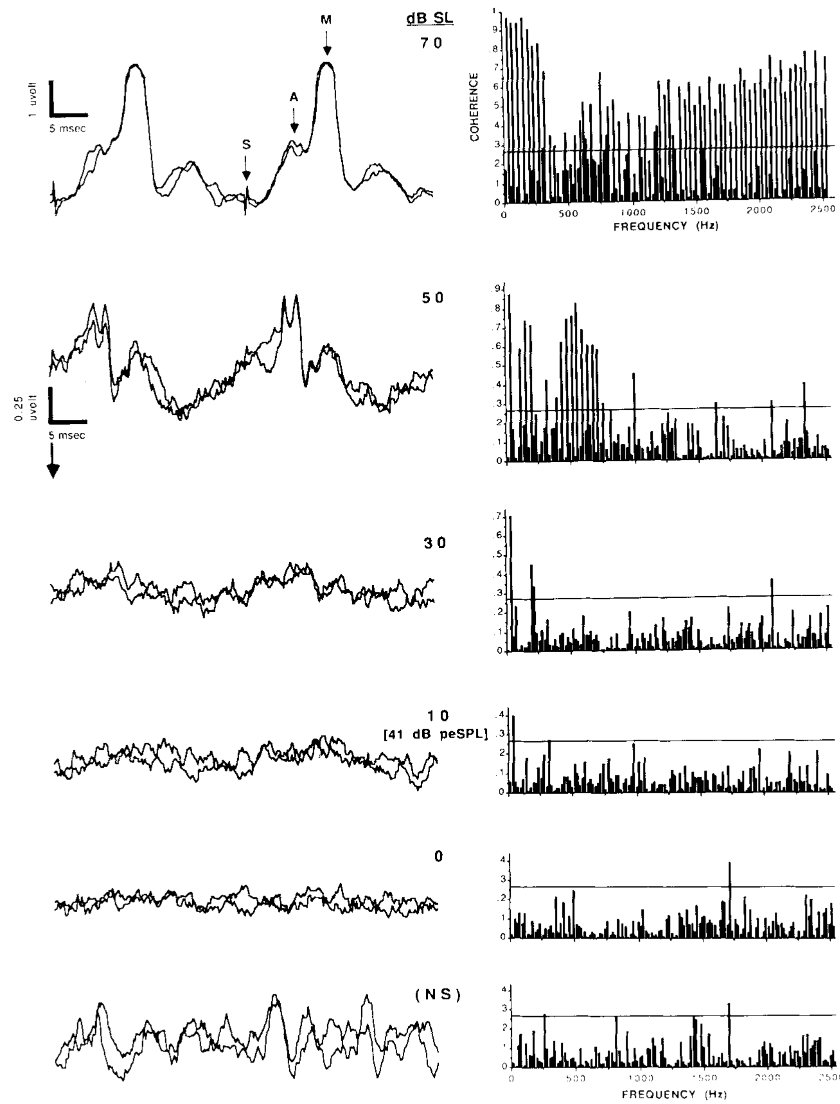

Figure 5. (*Left side*), Replicated scalp responses to alternating-polarity clicks. Intensity is indicated and decreases from top to bottom; NS, no stimulus; S, stimulus artifact; A, auditory brain stem response; M, myogenic response. (*Right side*) Corresponding coherence functions are shown (see text), with the critical value at the 0.01 level of significance.

sity, for two subjects. Significant coherence is present for all intensities down to 10 dB SL (equivalent to 41 and 44 dB peSPL for the two subjects), and is absent for 0 dB, −10 dB, and the no-stimulus condition.

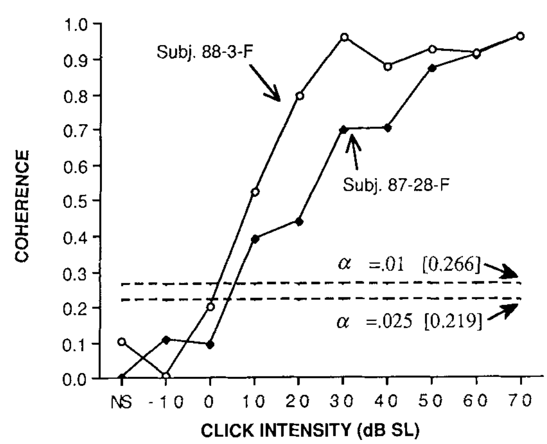

Figure 6. Forty Hz coherence (to alternating-polarity clicks) is shown for each of two subjects as a function of stimulus intensity. NS, no stimulus.

Effects of Varying q, n, and ℓ

Recall (equation 9) that coherences estimated without-subaveraging will always (and spuriously) equal 1. Increasing the number of subaverages (q), while holding constant the number of responses per subaverage (n), should reduce all coherence estimates toward their true values. As Figure 7 illustrates (using responses to 40 Hz alternating-polarity clicks at a low intensity of 40 dB SL), this results in improved identification of those frequencies genuinely contributing to the evoked potential. Note that the critical values (from Table 2) decrease as q increases.

Alternatively, one can spend data acquisition time by

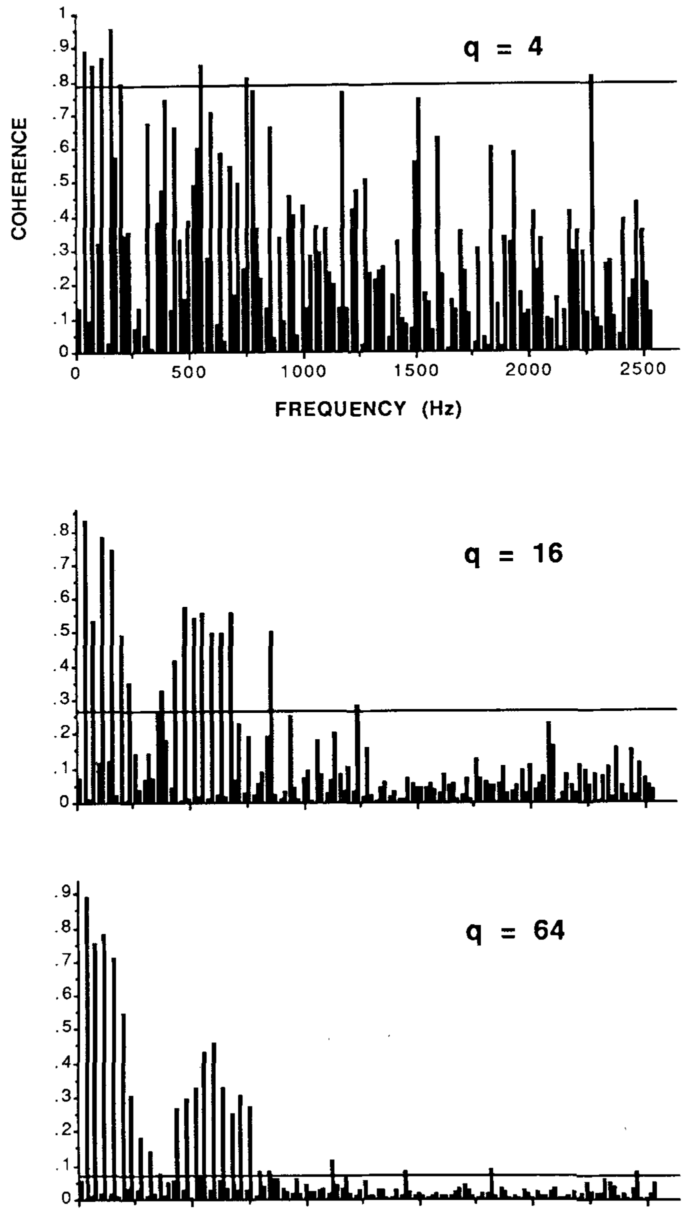

Figure 7. The effect of increasing the number of subaverages (q) on the coherence function is shown; the critical values at the 0.01 level of significance are included. Click intensity = 40 dB SL (74 dB peSPL).

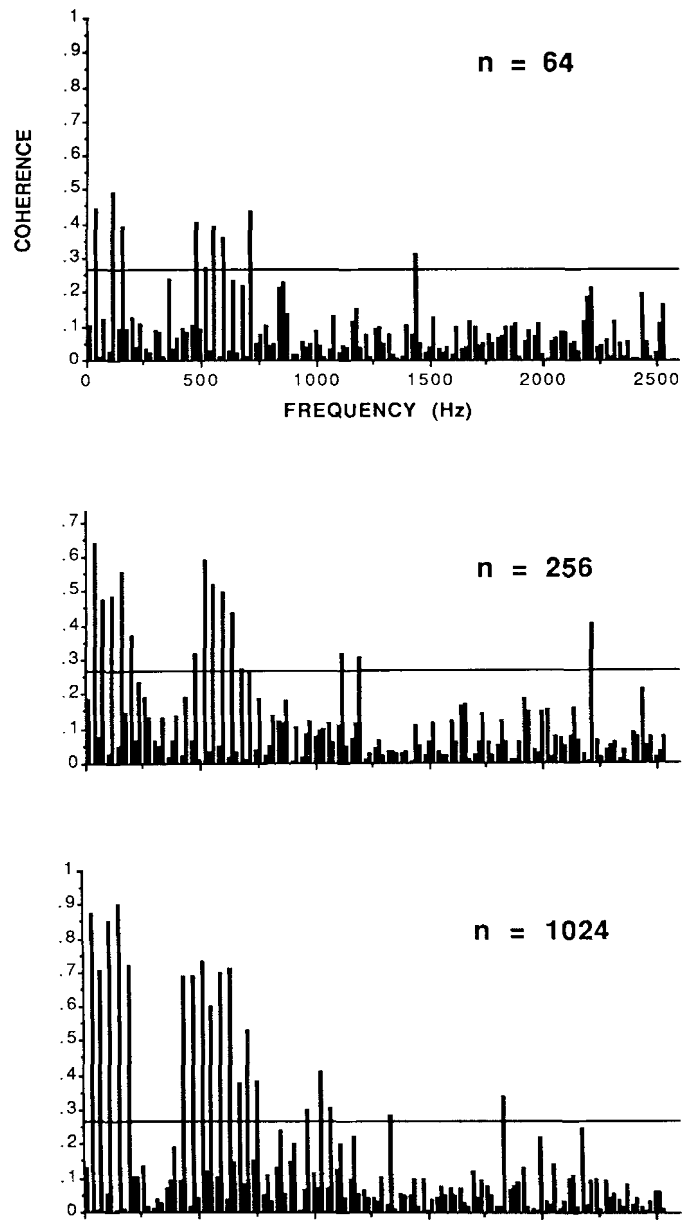

Figure 8. The effect of increasing the number of responses per subaverage (n) is shown; the critical value at the 0.01 level of significance is included. Click intensity = 40 dB SL (74 dB peSPL).

increasing n, leaving q constant. Since this increases S/N ratio, one expects (and finds) increasing coherence estimates, while critical values are unchanged (Fig. 8).

Increasing either q or n improves detection of significant coherence values and results in a greater number of contributing frequencies being identified, but both strategies require more time. One can improve coherence estimates without collecting more data by smoothing spectral estimates across frequency ($\ell > 1$). We initially tried this approach with alternating-polarity clicks. Since response to these stimuli, at moderate levels, shows significant coherence primarily for even harmonic frequencies, one would expect averaging across frequencies to be ineffective (unless the odd frequencies could be eliminated from calculations). This was indeed the case: for $\ell \geq 2$, all coherence estimates were below critical values. We then repeated the experiment with fixed-polarity clicks at 20/sec (Fig. 9). The coherence function ($\ell = 1$) is above the critical value to about 1100 Hz, and for increasing l, one finds both $\hat{\gamma}^2$ and critical values decreasing at about the same rate. At about 500 Hz, for example, $\hat{\gamma}^2$ is significant ($p < 0.01$) for $\ell = 1$ and $\ell = 4$, but not for $\ell = 16$.

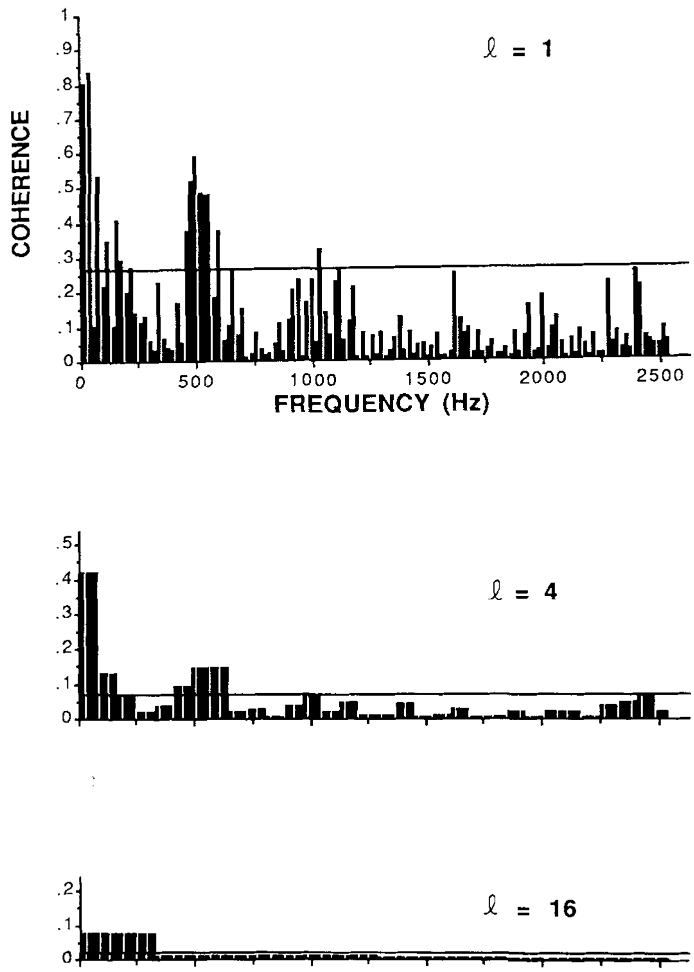

Figure 9. The effect of smoothing coherence estimates across frequency (ℓ > 1) is shown; the critical values at the 0.01 level of significance are included. Click intensity = 40 dB SL (71 dB peSPL).

DISCUSSION

Spectral Content

The magnitude-squared coherence function identifies those frequencies contributing significantly to an evoked potential. This information can be useful in specifying analog and/or digital filter parameters and sampling frequencies for improving evoked potential detection in time domain waveforms. For example, while significant coherence is present (disregarding stimulus artifact) up to about 1500 Hz for responses to high intensity 40-Hz clicks, low-intensity responses have a much narrower coherence spectrum. This suggests that different low-pass cutoffs might be selected for optimizing either latency measures (high intensity) or threshold estimation (low intensity).

Several investigators (e.g., Boston, 1981; Suzuki, Sakabe, & Miyashita, 1982; Laukli & Mair, 1981) have studied power spectra of human ABR; there is general agreement that the upper limit of response power is between 1500 and 2000 Hz for high-intensity clicks, with lower frequencies predominating for lower intensities.

Threshold Estimation

Coherence analysis provides an objective estimate of response threshold with a known risk of a false positive response, depending on the α value chosen. As the data in Figures 5 and 6 suggest, this method may prove to be more sensitive than simple inspection of replicated responses; we have shown this to be true for AMFR (DL Tucci, MJ Wilson, RA Dobie, unpublished data). Coherence estimates should be tested for significance against the critical values tabulated by Amos and Koopmans (1963), a few of which are reproduced in Table 2.

Envelope Detection

Responses to AM tones showed clear FFR to the stimulus frequencies. In addition, significant coherence was present for the modulating frequency (40 Hz), its 2nd and 3rd harmonics, and the frequencies 960, 1000, and 1040 Hz. Response energy at overtones of the envelope and component frequencies has been noted before for both 40 Hz (Kavanagh & Domico, 1986) and FFR (Davis & Britt, 1984; Snyder & Schreiner, 1984, 1987). Interpretation of these nonlinearities requires some discussion of the concept of an envelope. Our AM tone, although synthesized by addition of sinusoids, was identical to the product of a carrier (500 Hz) and a modulator (40 Hz plus DC). In one obvious sense, the latter is the envelope, and can be derived from the AM tone via the Hilbert transform (Thrane, 1984). However, the mechanism by which the envelope is extracted in the cochlea is basically half-wave rectification and low-pass filtering. This produces a more complex envelope whose strongest spectral components are precisely those best seen in the coherence function. Thus, it seems that scalp response to an intense 500 Hz AM tone consists primarily of FFR plus even-order nonlinearities equivalent to rectification.

Some subjects showed significant coherence for other frequencies: 420, 580, and 620 Hz. These could represent combination tones of the form $2f_1$-f_2 or $2f_2$-f_1.

Coherence around 1000 Hz was seen only at 60 to 70 dB SL, and 500 Hz coherence disappeared at about 40 dB SL. As no acoustic distortion was measurable around 1000 Hz (phone output more than 70 dB down re:500 Hz), and because FFR sensitivity declines monotonically for frequencies above 500 Hz, it is quite unlikely that the 1000 Hz responses were FFRs to acoustic distortion.

Choice of Analysis Parameters

As expected, increasing either q or n improves the detection of significant coherence; both increase the total number of responses averaged. For a fixed data collection period, q and n will be inversely proportional, and there is no obvious way to optimize their values. For AEPs, q between 16 and 64 seems to produce acceptable coherence functions, and we used q = 16 for most of the experiments in this paper because computational time increases with q.

Smoothing coherence estimates across groups of adjacent frequencies is relatively ineffective for ABR, probably because response phase changes rapidly relative to the spectral resolution (20 Hz) in these studies. Finer resolution, achieved by use of longer stimulus/response periods, could improve the situation, but at the expense of longer recording times.

Alternatively, one could average groups of coherence estimates for adjacent frequencies. This differs from the frequency smoothing previously described, which requires the averaging of complex cross-spectral density functions prior to calculation of a coherence estimate and depends on a minimal rate of change of phase in the system transfer function. Averages of coherence estimates for groups of m frequencies will have narrower distributions than the single estimates (standard error of mean = standard deviation $\div \sqrt{m}$), and critical values will be correspondingly lower. This approach could be useful in analyzing responses such as ABR.

CONCLUSIONS

Our goal has been to introduce and illustrate the magnitude-squared coherence function as an AEP analysis tool. This technique appears promising for purposes of objectively identifying stimulus-response relationships in the frequency domain. It could be useful for threshold estimation for a variety of evoked potentials. Whether it offers any advantages over the other frequency domain analytic approaches described in the "Calculation" section remains to be seen. γ^2 can be calculated on-line, and is computationally about as complex as "phase coherence" and less complex than the T^2 test. It has the potential advantage over both of permitting smoothing across frequencies to reduce variance without increasing averaging time.

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